



Basic Information:

Experiment No: 04

Date:

Title: Design a “farm animals” java application with the details of animals like cow, pig, horse. Consider the following details like where they stay, what they eat, the sound they make by using classes and objects.



Objective /Aim:

To develop a Java program using multiple classes to represent different farm animals such as Cow, Pig, and Horse, and display their place of living, food, and sound.



System Requirements:

Hardware:

- > Standard PC/Laptop
- > Intel Core i3/i5 processor or equivalent
- > Minimum 4 GB RAM
- > Minimum 500 MB free

Software:

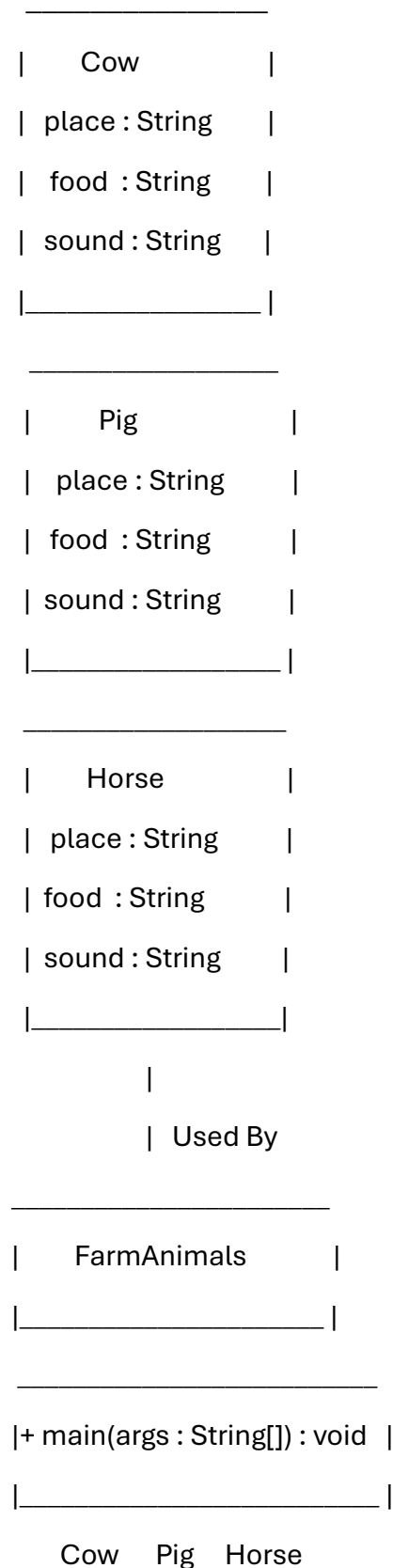
- > Operating System: Windows / Linux / macOS
- > Java Development Kit (JDK 17 or JDK 21)
- > Text Editor or IDE such as Eclipse, IntelliJ IDEA, NetBeans, or VS Code
- > Command Prompt / Terminal

Algorithm/Logic Description:

1. Start the program.
2. Create a Cow class with place, food, and sound.
3. Create a Pig class with place, food, and sound.
4. Create a Horse class with place, food, and sound.
5. Create objects for Cow, Pig, and Horse in the main() method.
6. Display the details of the Cow.
7. Display the details of the Pig.
8. Display the details of the Horse.

9. Stop the program.

UML Class Diagram:



Source Code:

```
package record;

// Class representing a Cow
class Cow {
    // Place where the cow stays
    String place = "Cowshed";
    // Food eaten by the cow
    String food = "Grass";
    // Sound made by the cow
    String sound = "Moo";
}

// Class representing a Pig
class Pig {
    // Place where the pig stays
    String place = "Pigsty";
    // Food eaten by the pig
    String food = "Vegetables";
    // Sound made by the pig
    String sound = "Oink";
}

// Class representing a Horse
class Horse {
    // Place where the horse stays
    String place = "Stable";
    // Food eaten by the horse
    String food = "Grass and hay";
    // Sound made by the horse
    String sound = "Neigh";
}
```

```
// Main class

public class FarmAnimals {

    public static void main(String[] args) {

        // Create an object of Cow class

        Cow cow = new Cow();

        // Create an object of Pig class

        Pig pig = new Pig();

        // Create an object of Horse class

        Horse horse = new Horse();

        // Display Cow details

        System.out.println("Cow");

        System.out.println("Stays: " + cow.place);

        System.out.println("Eats: " + cow.food);

        System.out.println("Sound: " + cow.sound);

        // Print a blank line

        System.out.println();

        // Display Pig details

        System.out.println("Pig");

        System.out.println("Stays: " + pig.place);

        System.out.println("Eats: " + pig.food);

        System.out.println("Sound: " + pig.sound);

        // Print a blank line

        System.out.println();

        // Display Horse details

        System.out.println("Horse");

        System.out.println("Stays: " + horse.place);

        System.out.println("Eats: " + horse.food);

        System.out.println("Sound: " + horse.sound);

    }

}
```

```
}
```

Relationship:

FarmAnimals —uses—> Cow

FarmAnimals —uses—> Pig

FarmAnimals —uses—> Horse

Output Screenshot:



```
Console X
-terminated- FarmAnimals [Java Application] C:\Program Files\Java\jdk-25.0.3\bin\java.exe [24-Sept-2025, 5:24:21 pm - 5:24:21 pm elapsed: 00:00] [pid: 22712]

Cow
Stays: Cowshed
Eats: Grass
Sound: Moo

Pig
Stays: Pigsty
Eats: Vegetables
Sound: Oink

Horse
Stays: Stable
Eats: Grass and hay
Sound: Neigh
```

Result:

The FarmAnimals Java program was successfully implemented and executed. The program creates objects for the Cow, Pig, and Horse classes and displays the place where each animal stays, the food it eats, and the sound it makes.